10. Let $x(t) = |\sin t|$.

(a) Plot the signals $x'(t)$ and $x''(t)$.

(b) Find the Fourier series coefficients of the signal $x(t) + x''(t)$.

(c) Use the differentiation property to find the Fourier series coefficients of the signal $x(t)$.

$x(t) = \sin t \quad t \in [0, \pi)$

$= -\sin t \quad t \in [\pi, 2\pi)$

$\Rightarrow x'(t) = \cos t \quad t \in (0, \pi)$

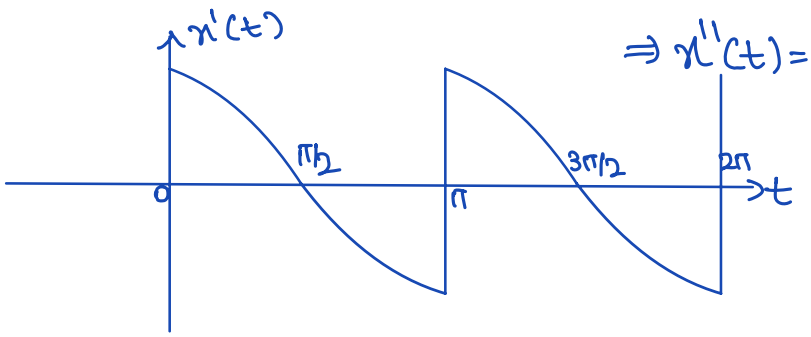
$= -\cos t \quad t \in (\pi, 2\pi)$

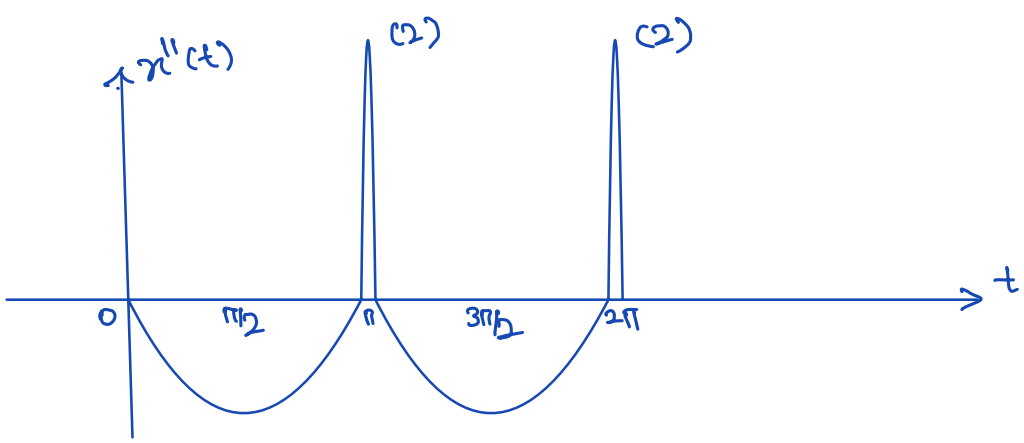
$\Rightarrow x''(t) = -\sin t \quad t \in (0, \pi)$

$\sin t \quad t \in (\pi, 2\pi)$

$\delta(t-\pi) \quad t = \pi$

(a)





both are periodic with fundamental period π

(b) $y(t) = x(t) + x''(t) = \sum_{n=-\infty}^{\infty} 2\delta(t - n\pi) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t}$

$a_k = \frac{2}{T} \int_{-T/2}^{T/2} y(t) e^{-jk\omega_0 t} dt \qquad T = \pi \quad \omega_0 = 2$

$a_k = \frac{2}{\pi} \int_{-\pi/2}^{\pi/2} \left(\sum_{n=-\infty}^{\infty} \delta(t - n\pi) \right) e^{-2jkt} dt \Rightarrow a_k = \frac{2}{\pi}$

(c) let $x(t) = \sum_{k=-\infty}^{\infty} P_k e^{jk\omega_0 t}$

$$x(t) + x''(t) = \sum_{k=-\infty}^{\infty} \left(P_k e^{jk\omega_0 t} + (\omega_0 j)^2 k^2 P_k e^{jk\omega_0 t} \right)$$

$$\sum_{k=-\infty}^{\infty} \frac{2}{\pi} e^{jk\omega_0 t} = \sum_{k=-\infty}^{\infty} \left[\left(P_k + (\omega_0 jk)^2 P_k \right) e^{jk\omega_0 t} \right]$$

$$\Rightarrow P_k \left[1 + (2)^2 k^2 (-1) \right] = 2/\pi$$

$$\Rightarrow \boxed{P_k = \frac{2}{\pi(1 - 4k^2)}}$$